

The empirical recurrence for OEIS A236647 is false at every index its b-file can test, and the correct one, of order 38

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Abstract

OEIS A236647 carries an empirical recurrence of order 34 contributed by R. H. Hardin. It is false. The entry's own b-file runs to $n = 210$, so the recurrence can be tested at 176 indices, and it fails at all 176; the first is $n = 35$, where the recurrence falls short of the b-file's 36-digit value by 145084608. A transfer matrix on 224 states, brute-force confirmed and reproducing all 210 b-file terms, gives the recurrence that does hold: it has order 38, its first 34 coefficients are exactly the published ones, and it holds for every $n \geq 40$, failing at $n = 39$ and there only. The published line is the correct recurrence with its last four terms dropped, and the entry's own parent table A236651 already says so, recording this column as "[order 38] for $n > 39$ ".

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1 The sequence and the claim

OEIS A236647 is "Number of $(n+1) \times (2+1)$ 0..2 arrays colored with the maximum plus the upper median minus the lower median minus the minimum of every 2×2 subblock.". It has offset 1, publishes 16 terms in DATA and a b-file of 210. It carries one formula line, an empirical recurrence of order 34:

$$\begin{aligned} a(n) = & 13a(n-2) + 184a(n-3) + 3506a(n-4) + 9110a(n-5) \\ & + 21587a(n-6) + 63605a(n-7) - 342128a(n-8) - 864643a(n-9) \\ & + 1140624a(n-10) + 294974a(n-11) + 739053a(n-12) + 15509266a(n-13) \\ & - 16660349a(n-14) - 37563062a(n-15) + 41074149a(n-16) - 39645704a(n-17) \\ & - 48556860a(n-18) + 140778823a(n-19) - 72899036a(n-20) - 67241570a(n-21) \\ & + 329687733a(n-22) - 46469782a(n-23) - 133466871a(n-24) + 319656720a(n-25) \\ & - 408928291a(n-26) - 40519358a(n-27) + 151099557a(n-28) - 160188966a(n-29) \\ & + 104521624a(n-30) - 63973040a(n-31) - 163603316a(n-32) + 759648a(n-33) \\ & + 12746672a(n-34). \quad (1) \end{aligned}$$

As of the live entry (revision 6, last modified Jul 23 2025) this is recorded as empirical, with no range stated and nothing recording it as settled or corrected.

With offset 1 and order 34 the line first asserts something at $n = 35$. DATA stops at $n = 16$, so on the DATA alone the claim is untestable — but the entry links a b-file running to $n = 210$, which tests it 176 times.

2 The published recurrence fails at all 176 indices its b-file can test

Substituting the b-file's own integers into the right-hand side of (1):

n	$a(n) - \text{RHS of (1)}$
35	145084608
36	1515557632
37	14366219776
38	133896336896
39	1253784855040
40	11713661278720

Proposition 1. (1) is false. It fails at every one of the 176 indices $n = 35, \dots, 210$ at which the entry's b-file can test it. The smallest is $n = 35$, where

$$a(35) = 470259440232820855339223326337397112$$

and the right-hand side of (1) gives

$$470259440232820855339223326192312504,$$

a difference of 145084608.

Proof. Direct evaluation in exact integer arithmetic on the 210 integers the entry's b-file publishes. No model enters this computation. $\square$

3 What the entry counts

The wording is Hardin's and is terse, so it is worth fixing precisely, and then checking the reading rather than assuming it. Let A be an $(n+1) \times 3$ array over $\{0, 1, 2\}$. Each 2×2 subblock has four entries; sorted as $w \leq x \leq y \leq z$, give the subblock the *colour*

$$\chi = z + y - x - w,$$

the maximum plus the upper median minus the lower median minus the minimum. The subblocks of A form an $n \times 2$ grid, and the condition is that *horizontally and vertically adjacent subblocks get different colours* — the array is properly coloured by χ . Let $a(n)$ count such arrays.

For $n = 1$ there are two subblocks, side by side, and the condition is one inequality; for $n = 2$ there are four, in a 2×2 grid. Direct enumeration over all $3^{3(n+1)}$ arrays gives

$$484, \quad 4500, \quad 41980 \quad (n = 1, 2, 3),$$

the entry's first three terms exactly.

4 The transfer matrix, and the recurrence that does hold

A row of the array is one of $3^3 = 27$ tuples, and whether a new row may be added depends on the previous row and on the colours of the subblock row just completed — on nothing else. Take as vertices the pairs (last row, colour pair), keep those reachable, and there are $S = 224$ of them; let M be the adjacency matrix of the legal extensions and $\mathbf{u}$ the distribution after the first two rows. Then $a(n) = \mathbf{1}^\top M^{n-1} \mathbf{u}$ for $n \geq 1$. Evaluated exactly, this reproduces all 16 published DATA terms and all 210 b-file terms, and its first three values are the brute-force counts above.

Being a walk count, a is C -finite. Solving over $\mathbf{Q}$ from the model's terms, the recurrence that holds on the tail has order 38 with integer coefficients:

$$\begin{aligned}
a(n) = & 13a(n-2) + 184a(n-3) + 3506a(n-4) + 9110a(n-5) \\
& + 21587a(n-6) + 63605a(n-7) - 342128a(n-8) - 864643a(n-9) \\
& + 1140624a(n-10) + 294974a(n-11) + 739053a(n-12) + 15509266a(n-13) \\
& - 16660349a(n-14) - 37563062a(n-15) + 41074149a(n-16) - 39645704a(n-17) \\
& - 48556860a(n-18) + 140778823a(n-19) - 72899036a(n-20) - 67241570a(n-21) \\
& + 329687733a(n-22) - 46469782a(n-23) - 133466871a(n-24) + 319656720a(n-25) \\
& - 408928291a(n-26) - 40519358a(n-27) + 151099557a(n-28) - 160188966a(n-29) \\
& + 104521624a(n-30) - 63973040a(n-31) - 163603316a(n-32) + 759648a(n-33) \\
& + 12746672a(n-34) \\
& 1733120a(n-35) + 12956032a(n-36) + 5889024a(n-37) - 243712a(n-38). \quad (2)
\end{aligned}$$

Its first 34 coefficients are, term for term, those of (1). The published line is (2) with its last four terms, $+1733120a(n-35) + 12956032a(n-36) + 5889024a(n-37) - 243712a(n-38)$, missing.

Theorem 2. (2) holds for every $n \geq 40$, and fails at $n = 39$, where the two sides differ by 114688.

Proof. Let $p(t)$ be the characteristic polynomial of (2),

$$\begin{aligned}
p(t) = & t^{38} - 13t^{36} - 184t^{35} - 3506t^{34} - 9110t^{33} - 21587t^{32} \\
& - 63605t^{31} + 342128t^{30} + 864643t^{29} - 1140624t^{28} - 294974t^{27} - 739053t^{26} \\
& - 15509266t^{25} + 16660349t^{24} + 37563062t^{23} - 41074149t^{22} + 39645704t^{21} + 48556860t^{20} \\
& - 140778823t^{19} + 72899036t^{18} + 67241570t^{17} - 329687733t^{16} + 46469782t^{15} + 133466871t^{14} \\
& - 319656720t^{13} + 408928291t^{12} + 40519358t^{11} - 151099557t^{10} + 160188966t^9 - 104521624t^8 \\
& + 63973040t^7 + 163603316t^6 - 759648t^5 - 12746672t^4 - 1733120t^3 - 12956032t^2 \\
& - 5889024t + 243712,
\end{aligned}$$

and put $w = p(M)\mathbf{u}$, formed by 38 matrix-vector products in exact integer arithmetic. For $n \geq 39$,

$$a(n) - \sum_{i=1}^{38} c_i a(n-i) = \mathbf{1}^\top M^{n-39} w.$$

At $n = 39$ this is $\mathbf{1}^\top w = 114688 \neq 0$, which is the stated exception. For $n \geq 40$ it is $\mathbf{1}^\top M^m(Mw)$ with $m = n-40 \geq 0$, and that vanishes for every $m = 0, \dots, S$; by the Cayley-Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so it vanishes for every $m \geq 0$. $\square$

Solving for the recurrence only *proposes* (2); what makes it a theorem is the annihilation test, which is independent of how the candidate was found.

Corollary 3. (1) fails for every $n \geq 35$, not merely at the 176 indices the *b*-file exhibits.

Proof. For $n \geq 40$, subtracting (1) from (2) makes a failure of (1) at n equivalent to

$$1733120a(n-35) + 12956032a(n-36) + 5889024a(n-37) - 243712a(n-38) \neq 0.$$

Every state of the transfer graph has out-degree at least 5, checked over all 224 of them, so every admissible array of $n+1$ rows extends to one of $n+2$ rows and a is nondecreasing; every $a(m)$ is a count of a nonempty set, so $a(m) \geq a(1) = 484 > 0$. Hence each of $a(n-35), a(n-36), a(n-37)$ is at least $a(n-38)$ and the expression is at least $(1733120 + 12956032 + 5889024 - 243712)a(n-38) > 0$. For $35 \leq n \leq 39$ the table in Section 2 exhibits the failure directly. $\square$

5 Verification

Five checks.

First, the disproof in Section 2 uses only the 210 integers of the entry’s own b-file. It does not depend on the model or on Theorem 2.

Second, the reading of the entry’s wording was not assumed. It was confirmed by direct enumeration over all $3^{3(n+1)}$ arrays for $n = 1, 2, 3$, which gives 484, 4500, 41980 — the entry’s first three terms. The same rule, applied at width 2 instead of 3, reproduces the first four terms of A236646, the $k = 1$ column of the same table.

Third, the transfer model reproduces all 16 published DATA terms and all 210 b-file terms, every one and not a sample.

Fourth, (2) was checked directly against the b-file at every index from $n = 39$ to $n = 210$: it fails at $n = 39$ and holds at all 171 others, matching Theorem 2 exactly. Every order $1, \dots, 37$ was solved for on the tail (fitted at $n = 140$) and every one fails there — order 37 at 22 of the remaining 61 indices, and the failure count rises steadily as the order falls — so 38 is the least order that holds eventually, and (1)’s 34 is not merely incomplete bookkeeping but too short to be repaired by any choice of coefficients.

Fifth, the annihilation test was carried to the full Cayley–Hamilton bound $m \leq S = 224$, not to a sampled prefix.

Remark 4. The entry’s parent table already carries the right answer. A236651 is the table of which A236647 is column 2, and its “Empirical for column k” block records

k=1: [linear recurrence of order 11]
k=2: [order 38] for n>39

— naming both the order, 38, and the exception, $n > 39$, that Theorem 2 proves. The column entry prints an order-34 line instead. Two statements on the OEIS about the same recurrence disagree, and the table is the one that is right.

This is the second instance of the same shape found by the same scan. The other is A269637, column 5 of A269640, where the table says “[order 13]” and the column entry publishes an order-10 line that its own DATA refutes ten times over. Across 6430 table/column pairs in the OEIS these two are the only disagreements, and both are the column entry’s line, truncated.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequences A236646, A236647 and A236651.
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